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EIT & MEC - Digi Lab 2: Sequential Circuits Simulation	Rev. 1.1	05/06.05.2026	Page 1 of 1

1. Models:

The file "Lab02_SRC_Simulation_files.zip" contains both the symbol files of the integrated circuits needed for the drawing the schematic and the library file 74hc.lib needed for simulating the circuit.

2. Hints:

- 2.1. Unfortunately, the IC 74HC160 (decade counter) is not available as LTSpice model. Please use the IC 74HC161 (binary counter) instead. The two ICs are identical (both pinout and functionality) with only one difference: while the first counts from 0 to 9, the latter counts from 0 to 15. The datasheet of 74HC161 can also be found in the directory "Lab Datasheet".
- 2.2. When adding the IC 74HC221 to the circuit, the following is needed: right-click on the part symbol in the schematic. The properties window of the part opens. The SpiceLine contains the following: "VCC=5 SPEED=1.0 TRIPDT=1e-9". The last parameter states the minimal time slice for the simulation. This causes the simulation to stop at 200 μ s and to continue very slowly, it would take ages to complete. The cause is probably an accuracy issue in the part's model. Please modify as follows: "TRIPDT=1e-1". This will lead to a reasonable result in reasonable time with minimal inaccuracy (the time delay is slightly shorter).
- 2.3. The symbol files of both 74HC08 (4x 2In-AND-Gate) and 74HC32 (4x 2In-OR-Gate) are also given to ensure a fast success. Please keep in mind that these parts must be substituted with a combination of the parts specified in the task description.
- 2.4. The figure below shows how you a signal looks like in LTSpice. You can insert a voltage, then right click on the voltage source, and choose "advanced" and then "pulse":

